

Executive Summary

Comparative Study of Speech-to-Text Models for Noisy Real-World Audio

Project Objective:

The objective of this study was to compare leading ASR models and identify the most suitable solution for a voice-based customer support system operating in noisy environments.

Models Compared:

- OpenAI Whisper Base
- Faster-Whisper Base
- Facebook Wav2Vec2 Base

All models were evaluated using the LibriSpeech test-clean dataset under identical experimental conditions.

Key Results:

Model	WER ↓	Time (sec) ↓	Memory (MB) ↓
Whisper Base	0.0569	1.300	3.46
Faster-Whisper Base	0.0643	2.363	0.91
Wav2Vec2 Base	0.0341	1.077	52.90

Observations:

- Wav2Vec2 achieved the highest transcription accuracy and fastest inference.
- Faster-Whisper demonstrated the lowest memory usage.
- Whisper Base provided the most balanced overall performance for production deployment.

Recommendation:

OpenAI Whisper Base is recommended for production due to its balanced performance and suitability for real-world speech.

Business Impact:

Enables accurate speech transcription, improving customer support automation and operational efficiency.

Future Work:

Future work includes evaluating the models on noisy real-world customer support datasets and fine-tuning the selected model to further improve transcription accuracy and deployment performance.